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Final Report

The Art of Scientific Computing: Complex Systems

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Chapter 1

Sandpile Model

This project is based on the "Complex Systems" project outline [5]. In this section, we introduce the sandpile model.

1.1 Self-Organised Criticality and Scale Invariance

The concept of *self-organised criticality* (SOC) was introduced in 1987 by Bak, Tang and Wiesenfeld [1, 2]. Self-organised criticality is a behaviour exhibited by certain complex systems which naturally evolve towards a particular critical state. Further, a system with SOC should develop the critical behaviour through its own dynamics, without needing to calibrate particular parameters.

An example of a critical system with SOC is given by the occurrence of earthquakes in a region. The *Gutenberg-Richter law* [4] describes how, for a given region, the number of earthquakes N of a magnitude M are related by

$$N = 10^{a-bM}$$

for some constants a and b. Since magnitude is a logarithmic measure of earthquake size (i.e., seismic energy E is related to magnitude by a relationship of the form $E \propto c^{dM}$ for some constants c and d), it follows that the number of earthquakes is a power law in the energy of the earthquake.

This power-law behaviour is typical of systems with SOC. More generally, systems with SOC typically display *scale invariance* in one form or another, meaning that some statistical properties remain similar when the scale of system changes.

Question 1. Why is it that a system exhibiting scale invariance will have phenomena which are described by a power law?

A power law has the property that the form of the distribution is unchanged by changing the scale. For instance, if we have a distribution of the form $f(x) \propto x^a$ for a constant a, and we scale x by a factor of λ, then $f(\lambda x) = \lambda^a f(x) \propto x^a$. So the resulting distribution of $f(\lambda x)$ is of the same form as that of $f(x)$.

1.2 Bak-Tang-Wiesenfeld Model

The Bak-Tang-Wiesenfeld (BTW) model is an easy-to-implement way we can observe SOC. It imagines a grid where grains of sand are dropped onto cells, stacking on top of each other until they reach a particular "height" (the *stability threshold*), at which point the grains of sand "topple" onto the adjacent cells. The topples can result in an adjacent cell reaching the stability threshold, which then causes that cell to topple. We call such a sequence of topples an "avalanche". It is through the statistics of these avalanches which we observe SOC.

We now describe the formal construction of the BTW model. We define a subset* $L \subseteq \mathbb{Z}^2$ which we will call the "sandpile", and a function $p_t : L \to \mathbb{Z}_{\geq 0}$ for each time point $t \in \mathbb{Z}_{\geq 0}$. We call an element of L a *site* in the sandpile, and the function $p_t(x)$ is the number of grains of sand at the site x at time t. We set $p_0(x) = 0$ for every $x \in L$. At each time point t, a site $x \in L$ is sampled uniformly and we set $p_{t+1}(x) = p_t(x) + 1$. Then, if any site $y \in L$ has $p_{t+1}(y) \geq k$, where $k = 4$ is the stability threshold, we decrease $p_{t+1}(y)$ by k and increase $p_{t+1}(z)$ by 1 for all $z \in L$ such that $|y - z| = 1$. This process is repeated until $p_{t+1}(y) < k$ for all $y \in L$.

Question 2. In what ways does the BTW model satisfy the properties required for a system to exhibit SOC?

The BTW model exhibits SOC because it can develop the critical behaviour through its own dynamics, without needing to calibrate particular parameters. It is self-organising, because the repeated avalanches and addition of grains bring the system into a critical state. The fact that most of the grains are conserved in the lattice allows the system reach and maintain the critical state, and in this critical state, the avalanches can occur over different scales.

Further, the model is slow driving, since the grains being dropped are small localised changes to the system. If we add a grain of sand to the pile, the sand pile can either remain stable or produce an avalanche. The microscopic phenomena is the same in both cases. The size of an avalanche depends on the configuration of the pile, not just the initial disturbance by the dropped grain of sand.

Question 3. Why is it important for the lattice to have finite boundaries? What if we had an infinite lattice, but dropped the sand on only a (finite) subset of the lattice?

It is necessary to have finite boundaries so that sand can be dissipated, allowing the pile to reach a statistically stationary state. If we had a finite lattice but the lattice "wraps around", then the total amount of sand would strictly increase every time a grain is added. As a consequence, the pile would eventually have so many grains that it can never reach a stable state. If we had an infinite lattice but only dropped sand on a finite subset, then the avalanche statistics would change as the number of grains outside the finite subset grows.

*In our implementation, L will be a subset of the form $[1, \ldots, N] \times [1, \ldots, N]$ for some $N \in \mathbb{Z}_{\geq 1}$. Further, the model can easily be generalised to consider subsets of $\mathbb{Z}^d$ for any $d \geq 1$.

Question 4. We can define the average height of each pile to be

$$p_t \coloneqq \frac{1}{NM} \sum_{i,j} p_t(i,j)$$

Without simulating anything, what might you expect this to be once the system has reached a steady state (if it ever does so)?

We can say, for certain that the average height in a stable state can be at most 3. If the average height is more than 3, then (by the pigeonhole principle) at least one site must be unstable. On one hand, if the average height were close to 3, then it is unlikely that every site would have a height less than 4. On the other hand, if the average height is too low then it is unlikely that the pile has reached the critical state, since grains of sand can only be lost at the boundary. Caracciolo and Sportiello [3] proved that the average height approaches 2.125 in the limit as the grid size increases, so we expect the average height to be smaller than this, increasing as the grid size increases.

Question 5. Explain the difference between: a steady state, a statistically stationary state, and an equilibrium state. Do you expect the btw model reach an equilibrium state or statistically stationary state? In terms of system observables, how might we characterise this state?

A steady state is where the behaviour of a system no longer changes over time. For the BTW model, this refers to the period once the transient effects of the grid starting empty disappear. A statistically stationary state is where the statistical properties of a system stay the same, while an equilibrium state is where the system has no external driving.

The BTW model will reach a statistically stationary state and not an equilibrium state. In particular, it will reach a state where the average number of grains lost will match the number of grains added, and at this point the system will be in a statistically stationary state. The dropping of grains is an external driving which will cause the system to never enter an equilibrium state.

We can characterise the state through the various statistics we can measure. The long-time average height will stabilise and the avalanche statistics will have a fixed distribution. We can also characterise it by when the net change in the number of grains over a period of time becomes 0.

Question 6. Is there a difference between sampling a system over time and sampling an ensemble of systems? If so, under what circumstances?

A system is ergodic if there exists a limiting distribution of its state, and the limit is independent of its current state. This means that the system can reach any state it should be able to reach, rather than being trapped in a subset of possible states. Sampling a system over time and sampling an ensemble of systems is fundamentally different, however, if the system is ergodic then the two methods are equivalent.

The BTW model is ergodic, since the random grain additions allow the system to enter any of its recurrent configurations. Consequently, we will sample the system over time rather than sampling an ensemble of systems, as it is more computationally

feasible to only have to get past the transient period of the system a limited number of times.

Question 7. Do you expect any correlation between the four properties we are recording for the avalanche? Justify why/why not.

We define an avalanche to be the sequence of topples that occur between the pile entering an unstable state after a grain has been dropped and the pile returning to a stable state. The four properties of an avalanche which we are recording are as follows:

Topples The number of topples in the avalanche;

Area The number of unique sites that were toppled in the avalanche;

Loss The number of grains which were deleted from the system by toppling off the lattice in the avalanche; and

Length The maximum distance between two sites which toppled in the avalanche.

We use the maximum distance between topples as our measure of length because it best describes the size of the avalanche (compared to, for instance, measuring the maximum distance between a topple and the site of the grain being dropped). We use the L^2 norm, since this is a more realistic measurement of distance.

We expect all four properties to be pairwise correlated because, for any of these properties, each additional topple that occurs in an avalanche either leaves the property the same or increases it. Roughly speaking, each property increases as the size of the avalanche increases.

1.3 Abelian Property

In this section, we prove that the topples of the BTW model are *abelian*. This means that in an unstable state, the order in which the unstable sites are toppled does not change the final stable configuration.

Let $L \subseteq \mathbb{Z}^d$ be the *sand pile lattice*, where $d \geq 1$ is the dimension of the lattice. Corresponding to a configuration of the sand pile p, we define a function $p : L \to \mathbb{Z}_{\geq 0}$, where $p(x)$ is the number of grains of sand on the lattice point x in the configuration p. Define k to be the *stability threshold* (i.e., the pile topples when a lattice point is of height at least k). Then p is a *stable* configuration if $p(x) < k$ for all $x \in L$, and it is *unstable* if $p(x) \geq k$ for at least one $x \in L$.

We define the *toppling matrix* Δ, which has rows and columns indexed by L. For $x, y \in L$, the entry $\Delta(x, y)$ describes the change in height of site y when there is a toppling at site x. The matrix Δ must satisfy the following conditions:

$$\forall x, y \in L \text{ where } x \neq y, \quad \Delta(x, y) = \Delta(y, x) \geq 0 \tag{1.1}$$

$$\forall x \in L, \quad \Delta(x, x) < 0 \tag{1.2}$$

$$\forall x \in L, \quad \sum_{y \in L} \Delta(x, y) \leq 0 \tag{1.3}$$

$$\sum_{x,y \in L} \Delta(x, y) < 0 \tag{1.4}$$

Question 8. For Equation 1.1 to 1.4, explain the significance and justify its necessity.

Equation 1.1 demands that when a site topples, no other site loses any grains. This is consistent with how we defined toppling. Equation 1.1 also insists that there is some symmetry in how toppling works, in the sense that the number of grains that go to a site y when x is toppled should be the same as the number of grains that go to x when y is toppled.

Equation 1.2 means that a site should lose grains when it is toppled. If Equation 1.2 does not hold for a site x, then when the height at x reaches k (which it eventually will do with probability 1), the pile will never be able to reach a stable configuration.

Equation 1.3 can be written as $-\Delta(x, x) \geq \sum_{y \in L \setminus \{x\}} \Delta(x, y)$ for all $x \in L$, meaning that when a site x is toppled, the number of grains gained by other sites is no more than the number of grains which are lost by x (i.e., we are not creating new grains when toppling).

Equation 1.4 demands that there is at least one site where toppling will result in at least one grain leaving the pile. If Equation 1.4 does not hold, the pile will end up enough grains that it can never reach a stable configuration.

In the BTW model, the toppling matrix is defined as follows:

$$\Delta_{\text{BTW}}(x, y) = \begin{cases} -2d & x = y \\ 1 & x \text{ and } y \text{ are nearest neighbours} \\ 0 & \text{else} \end{cases}$$

Question 9. Explain and justify the definition of the toppling matrix in the btw model and verify that it satisfies Equation 1.1 to 1.4. What is the stability threshold k?

The BTW toppling matrix Δ_{BTW} removes $2d$ grains from a toppled site x and adds 1 grain to each nearest neighbour of x. It is also a symmetric matrix, so it clearly satisfies Equation 1.1 and 1.2. Each site in the interior of the pile has $2d$ nearest neighbours, and each site on the boundary of the pile has less than $2d$ nearest neighbours. Therefore for each $x \in L$,

$$\begin{aligned} \sum_{y \in L} \Delta_{\text{BTW}}(x, y) &= \Delta_{\text{BTW}}(x, y) + \sum_{y \in L \setminus \{x\}} \Delta_{\text{BTW}}(x, y) \\ &= -2d + \#\{\text{nearest neighbours of x}\} \\ &\leq 0 \end{aligned}$$

Hence Δ_{BTW} also satisfies Equation 1.3. Since at least one site is on the boundary of the pile, we have

$$\sum_{x,y\in L} \Delta_{\text{BTW}}(x,y) = \sum_{x\notin\partial L} \sum_{y\in L\setminus\{x\}} \Delta_{\text{BTW}}(x,y) + \sum_{x\in\partial L} \sum_{y\in L\setminus\{x\}} \Delta_{\text{BTW}}(x,y) < 0$$

so Δ_{BTW} also satisfies Equation 1.4. The stability threshold is $k = 2d$.

For each $x \in L$, we then define the toppling operator T_x, which maps a configuration p to a new configuration $T_x p$, by

$$(T_x p)(y) = \begin{cases} p(y) + \Delta_{\text{BTW}}(x,y) & p(x) \geq k \\ p(y) & \text{else} \end{cases}$$

Question 10. Show that T_x and $T_{x'}$ commute for an unstable configuration p.

Proof. Let $x, x' \in L$ be distinct unstable sites and let $y \in L$ be any site. Then $p(x) \geq k$ and $p(x') \geq k$. It follows that for any $y \in L$, we have

$$\begin{aligned} (T_{x'}p)(y) &= p(y) + \Delta_{\text{BTW}}(x',y) \\ (T_{x'}p)(x) &= p(x) + \Delta_{\text{BTW}}(x',x) \\ (T_x p)(y) &= p(y) + \Delta_{\text{BTW}}(x,y) \\ (T_x p)(x') &= p(x') + \Delta_{\text{BTW}}(x,x') \end{aligned}$$

Since $x \neq x'$, and $p(x)$ and $p(x')$ are both at least k, it follows that $p(x) + \Delta_{\text{BTW}}(x',x) \geq k$ and $p(x') + \Delta_{\text{BTW}}(x,x') \geq k$. Therefore,

$$\begin{aligned} (T_x T_{x'} p)(y) &= \begin{cases} (T_{x'}p)(y) + \Delta_{\text{BTW}}(x,y) & (T_{x'}p)(x) \geq k \\ (T_{x'}p)(y) & \text{else} \end{cases} \\ &= (T_{x'}p)(y) + \Delta_{\text{BTW}}(x,y) \\ &= p(y) + \Delta_{\text{BTW}}(x',y) + \Delta_{\text{BTW}}(x,y) \\ &= (T_{x'}T_x p)(y) \end{aligned}$$

□

Question 11. Show that T is well defined. In other words, show that given some unstable configuration $p, p' = \mathbb{T}p$ is unique (you can't end with two different stable configurations). This is equivalent to the sequence of T_{x_i} required is unique up to re-ordering.

Suppose that p' and p'' are distinct configurations for $\mathbb{T}p$, and let A and B be the corresponding sequence of topplings. Since they start from the same stable configuration p, there is a site where the number of topples in one sequence is more than the number of topples in the other. Let $a(x)$ denote the number of topples at site x in A, and $b(x)$ denote the number of topples at site x in B. Without loss of generality, suppose $a(x) > b(x)$ for some $x \in L$.

Let x_0 be the site of the first toppling in A that causes the number of topplings in any site to be greater than the total number of topplings at that site in B. Let $a_0(x)$

denote the number of topplings in A at site x before this toppling. So $a_0(x_0) = b(x_0)$ and $a_0(x) \leq b(x)$ for all $x \in L$. If we denote the set of nearest neighbours of x_0 by I, then the height before this this toppling at x_0 is given by

$$\begin{aligned} & p(x_0) - 4a_0(x_0) + \sum_{x \in I} a_0(x) \\ \leq & p(x_0) - 4b(x_0) + \sum_{x \in I} b(x) \\ = & p''(x_0) \end{aligned}$$

This value is at most 3, since p'' is a stable configuration, contradicting that x_0 is the next topple in A. We conclude that $p' = p''$.

1.4 Aim

The aim of this project is to study SOC within the non-Gaussian statistics of the sandpile model. We will implement the basic BTW model and determine whether the statistics display scale invariance. Then, we will extend the BTW model and see how the behaviour of the system changes as a larger number of grains of sand are dropped at once.

Chapter 2

Implementation of the model

In this section, we describe the implementation of the sandpile model. The full code can be found in the file *sandpile_project*.

2.1 Test Model

We began by building a test model. This was broken up into four steps. The first was to create an array for the lattice and build a function which could drop grains of sand onto the lattice.

Listing 2.1: Sand dropping function

```
def random_drop_sand(pile, rng):
    """
    Drops a grain onto the pile at a random site.
```

Then we implement a function which can topple the pile at a particular site.

Listing 2.2: Toppling function

```
def topple(pile, unstable_site):
    """
    Applies the basic toppling rule to the pile, removing 4
       grains from an unstable site and 1 to each adjacent
       site.
```

Next, a function which can take an unstable pile and return it to a stable state.

Listing 2.3: Avalanche function

```
def avalanche(pile):
    """
    Repeatedly applies the basic toppling rule to the pile
       until it is stable (i.e., an avalanche). For the
       avalanche, we record the number of topples, the area,
       the total loss, and the length.
```

We then want to test the avalanche function to make sure it behaves as expected. We input some test piles to see if it correctly outputs the final grid and avalanche statistics.

```
Test Sandpile 1:
[[3 3 3]
 [3 4 3]
 [3 3 3]]
After avalanche:
[[1 3 1]
 [3 0 3]
 [1 3 1]]
Topples:  10  Area:  9  Total Loss:  12  Length:  2.8284271247461903

Test Sandpile 2:
[[4 3 3 3 3]
 [3 0 3 0 3]
 [3 3 0 0 3]
 [0 0 0 0 3]
 [0 0 0 0 0]]
After avalanche:
[[2 2 2 1 1]
 [2 0 1 3 1]
 [1 1 2 1 1]
 [1 1 0 1 0]
 [0 0 0 0 1]]
Topples:  13  Area:  13  Total Loss:  12  Length:  5.0
```

With these small grids, it is easy to check by hand that the final configurations and the statistics are correctly output by the avalanche function. We are now ready to test-run the simulation of the BTW model.

Listing 2.4: Test simulation function

```
1 def test_simulation(height, width, time_limit, rng):
2     """
3     Simulates the sandpile model for a given number of time
          steps.
```

2.2 Base Model

Having identified several computational inefficiencies in the test model, we now implement the BTW model in earnest. Firstly, we create a more efficient version of the avalanche function.

Listing 2.5: Optimised avalanche function

```
1 def fast_avalanche(pile, initial_sites):
2     """
3     A faster version of the avalanche function. Takes a pile
          and an initial unstable site and applies the
4     basic toppling rule to the pile until it is stable (i.e.,
          an avalanche).
```

Then, a more efficient version of the function which runs a simulation.

Listing 2.6: Optimised simulation function

```
def simulation(height, width, time_limit, rng, burnin=0):
    """
    Simulates the sandpile model for a given number of time
       steps.
```

We will run a simulation for a 10x10 grid, a 20x20 grid, and a 50x50 grid for 400,000 time steps. We will have burn-in periods of 2,000, 8,000, and 50,000 time steps respectively (equivalent to 20 times the number of sites in the grid).

Given that we expect the data may show power law behaviour, we define a function that makes a frequency scatter plot on log-log axes. It also fits a linear regression to the part of the data which is linear (after having taken the logarithm of the data values and the frequency). The function determines which part of the data is linear by finding the where the R^2 value of a linear regression drops below a certain threshold.

Listing 2.7: A graph-making function

```
def make_graph(data, title):
    """
    Prepares a log-log frequency scatter plot of the data,
       and fits a linear regression to the linear part of the
        data.
```

2.3 Extension

We now wish to compare how the avalanche statistics vary when multiple grains are dropped simultaneously at each time step. For this, we implement a new version of the simulation function with a parameter that allows a varying number of grains to be dropped.

Listing 2.8: A graph-making function

```
def simulation_multigrain(height, width, time_limit, rng,
   burnin=0, grains_per_drop=1):
    """
    Simulates the sandpile model where multiple grains are
       dropped at each time step.
```

We then run simulations where we add 1, 2, 5, and 10 grains at each time step to a 20x20 grid for 100,000 time steps with a burn-in period of 8,000 time steps.

Chapter 3

Analysis

3.1 Analysis of the base model

Question 12. Record the frequency with which avalanches have a specific number of topples, area, loss and length. What kind of distribution do you get for each? If possible, can you find a fit for the distribution?

We start by graphing each statistic against frequency on log-log axes, to see if they follow a power law. In Figure 3.1, we see that the number of topples follows a power law. That is, if t is the number of topples, then our data suggests $p(t) \propto t^{-a}$ where $a = 1.168$ for the 10x10 grid, $a = 1.183$ for the 20x20 grid, and $a = 1.096$ for the 50x50 grid. The values of a for the 10x10 grid and the 20x20 grid are close together, and the value of a for the 50x50 grid is slightly further away.

In Figure 3.2, we see that area also follows a power law, with $a = 1.159$ for the 10x10 grid, $a = 1.163$ for the 20x20 grid, and $a = 1.078$ for the 50x50 grid. Again, the values of a for the 10x10 grid and the 20x20 grid are close together, and the value of a for the 50x50 grid is slightly further away.

From Figure 3.3, it is clear that the loss does not follow a power law. We instead, in Figure 3.4, graph the loss again the logarithm of frequency. The data appears to be linear on this graph, except for on the larger size grid at low loss values. This indicates that loss may follow an exponential law. If l denotes loss, then our fit indicates that $p(l) \propto 10^{-al}$ where $a = 0.146$ for the 10x10 grid, $a = 0.106$ for the 20x20 grid, and $a = 0.058$ for the 50x50 grid.

In Figure 3.5, we see that the length of avalanches follows the trend of a power law, but we also notice that the graph is not monotonic. Between grid sizes, we see a consistent pattern of this behaviour. For instance, it is more likely for an avalanche to have a length of $\sqrt{5}$ than 2. It seems like this is an artefact of us using a lattice model and the way we measure length.

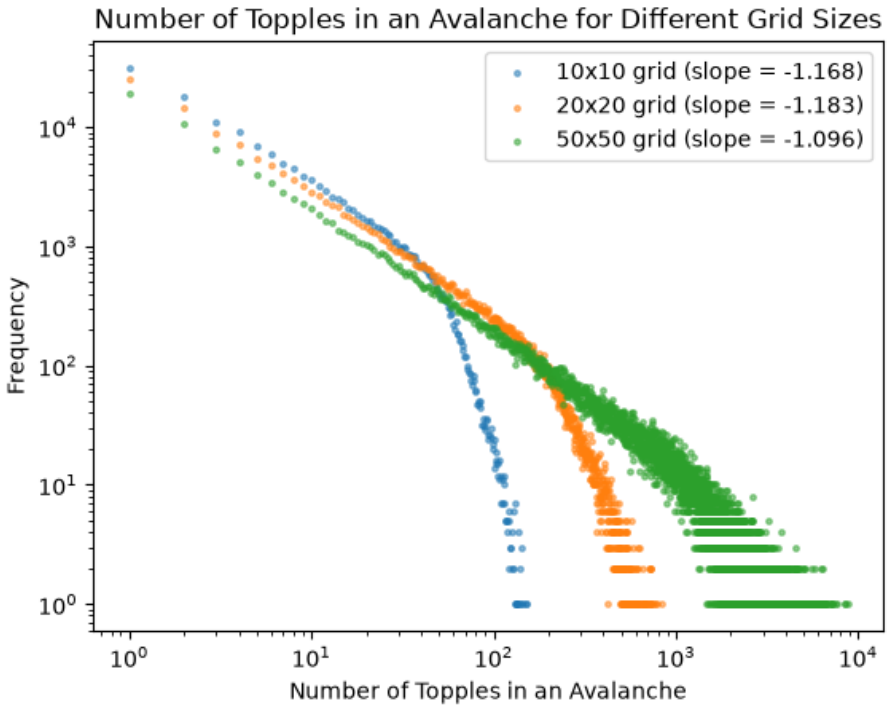


Figure 3.1: The number of topples for the base model. The graph has a linear section, so we do observe scale-invariance in the number of topples.

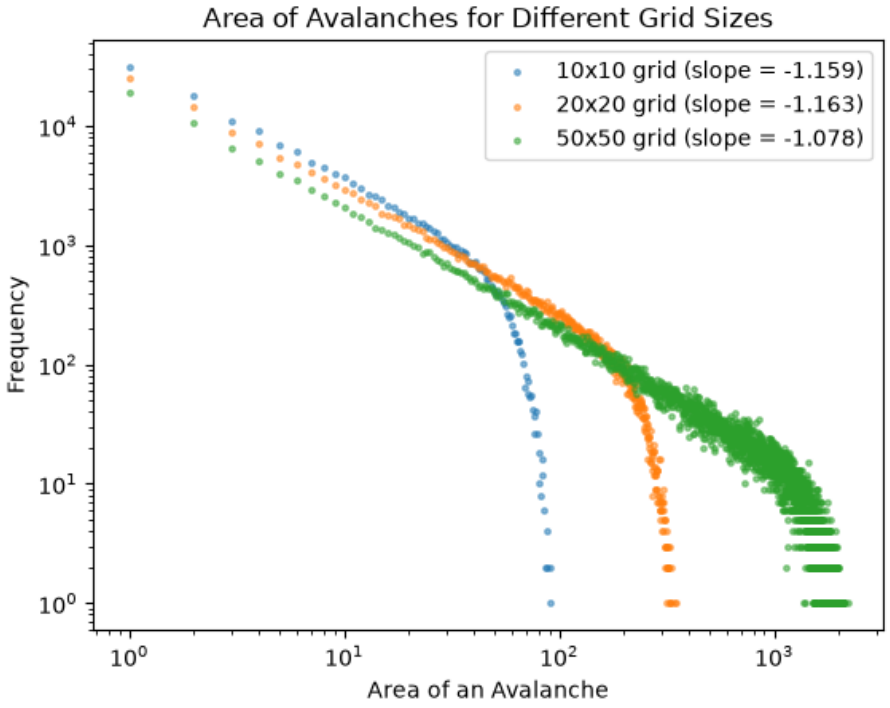


Figure 3.2: The area of avalanches for the base model. The graph has a linear section, indicating that there is scale-invariance in the area of avalanches.

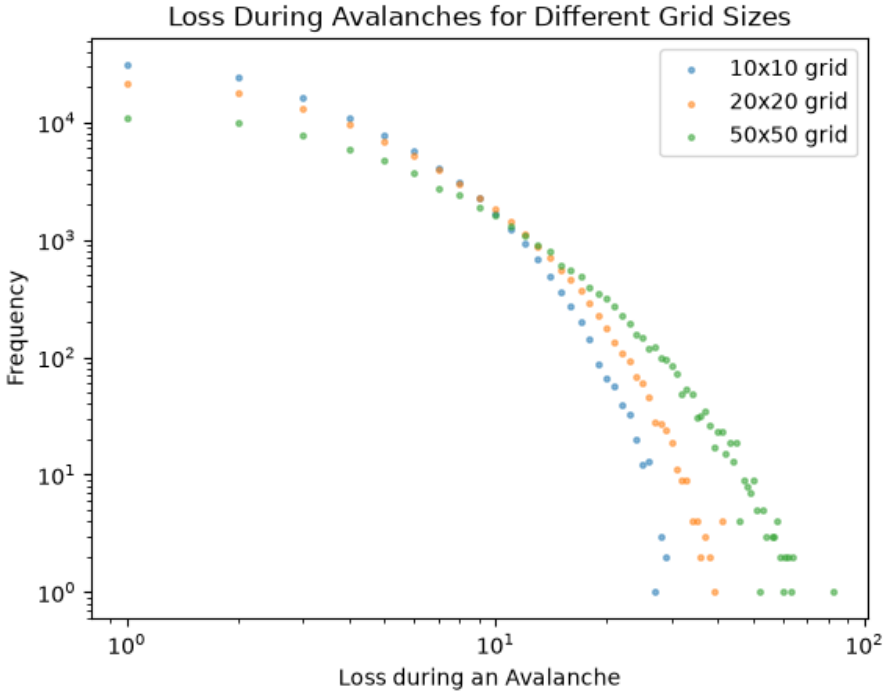


Figure 3.3: The loss for the base model. The graph is non-linear, indicating that loss does not follow a power law.

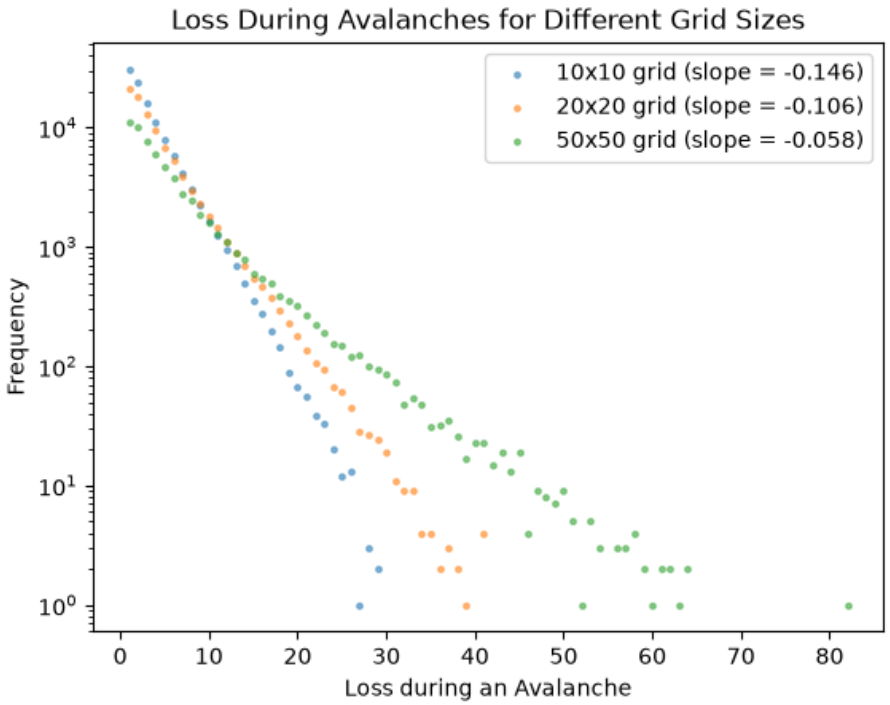


Figure 3.4: The loss for the base model, now with only the frequency axis being in logarithm. The graph has a linear section, indicating that loss may follow an exponential relationship.

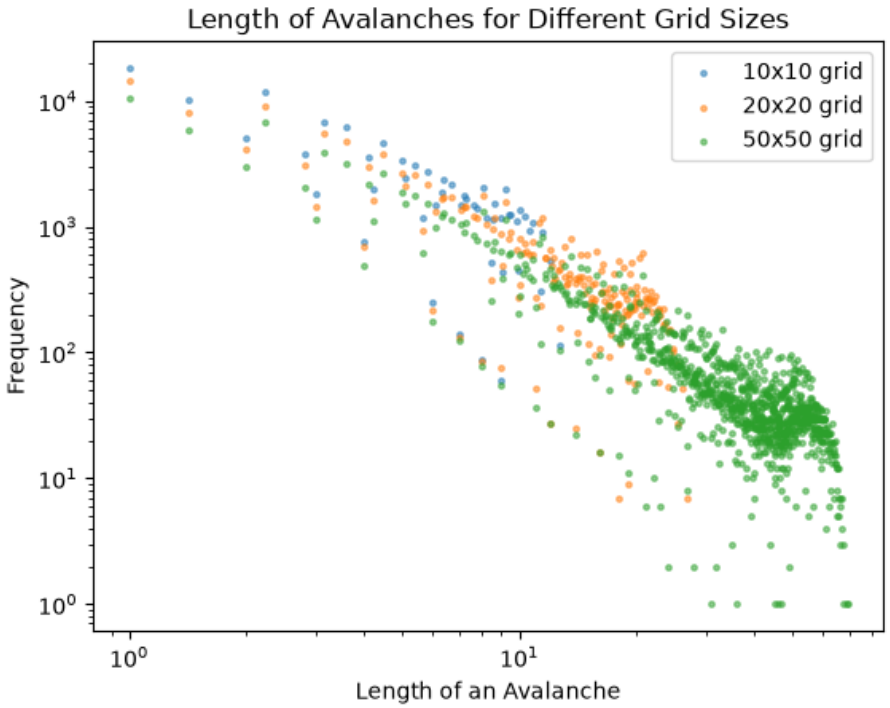


Figure 3.5: The length of avalanches for the base model. The graph is linear but with a lot of variance.

Question 13. Verify whether there was any correlation between the four properties we are recording for the avalanche.

In Table 3.1, we see that there was, in fact, positive correlation between each pair of statistics for each size grid. The correlation is strongest between topples and area, and weakest between length and loss. We see the strength of the correlation generally decreases as grid size increases, with the correlation between topples as loss being an exception to this as it increases from the 20x20 grid to the 50x50 grid..

Question 14. Compare your initial expectation for the average height to what your simulation does.

We measure the average height of the pile over 100,000 time steps (with a burn-in period) for a 10x10 grid, a 20x20 grid, and a 50x50 grid.

```
The average height for a 10x10 grid is: 2.006391836734694
The average height for a 20x20 grid is: 2.061899456521739
The average height for a 50x50 grid is: 2.097251712
```

We see that the average height ranges from 2.01 to 2.10 depending on the grid size, with a larger grid having a larger average height. This agrees with our hypothesis. Simulating larger grids would be required to get closer to the limit of 2.125 given by Caracciolo and Sportiello [3].

Table 3.1: Correlation matrices of avalanche statistics.

	Topples	Area	Loss	Length
		10×10 grid		
Topples	1.000000	0.985934	0.768923	0.897798
Area	0.985934	1.000000	0.774378	0.929118
Loss	0.768923	0.774378	1.000000	0.714528
Length	0.897798	0.929118	0.714528	1.000000
		20×20 grid		
Topples	1.000000	0.968383	0.732303	0.860467
Area	0.968383	1.000000	0.734632	0.923382
Loss	0.732303	0.734632	1.000000	0.674210
Length	0.860467	0.923382	0.674210	1.000000
		50×50 grid		
Topples	1.000000	0.936545	0.749283	0.802449
Area	0.936545	1.000000	0.727887	0.916770
Loss	0.749283	0.727887	1.000000	0.650059
Length	0.802449	0.916770	0.650059	1.000000

3.2 Analysis of the extended model

We now consider how changing the number of grains dropped at each time period changes the statistics. In Figure 3.6, we see that the number of topples still maintains this power law relationship. Further, the gradient increases as the number of grains increases, possibly approaching a constant relationship. We note that our slope-fitting function fails to fit the case when 10 grains are being dropped. The issue is with using the R^2 value to fit a function approaching a constant. If we consider how R^2 is defined,

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

it is clear that as the data approaches a constant, an R^2 value will approach 0. To fit a line in this case, we would need to use another method of measuring fit. For the area, we see a similar pattern in Figure 3.6.

We again use an exponential fit in Figure 3.8 for the loss statistic. As the number of grains increases, it appears that the relationship does not follow an exponential for small values of loss. As we have defined it, the length of an avalanche, seen in Figure 3.9, does not follow a power law for larger numbers of dropped grains. This is because grains dropped far apart on the lattice can result in an avalanche which starts in multiple places, making the length values artificially large.

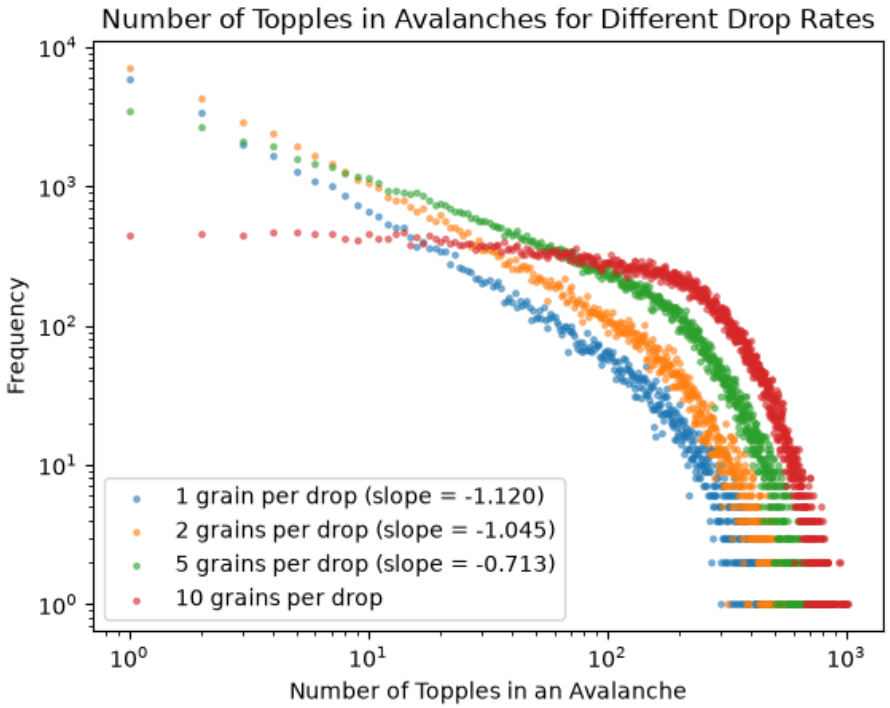


Figure 3.6: The number of topples for the extension model. The gradients of the graphs increase as the number of grains increases.

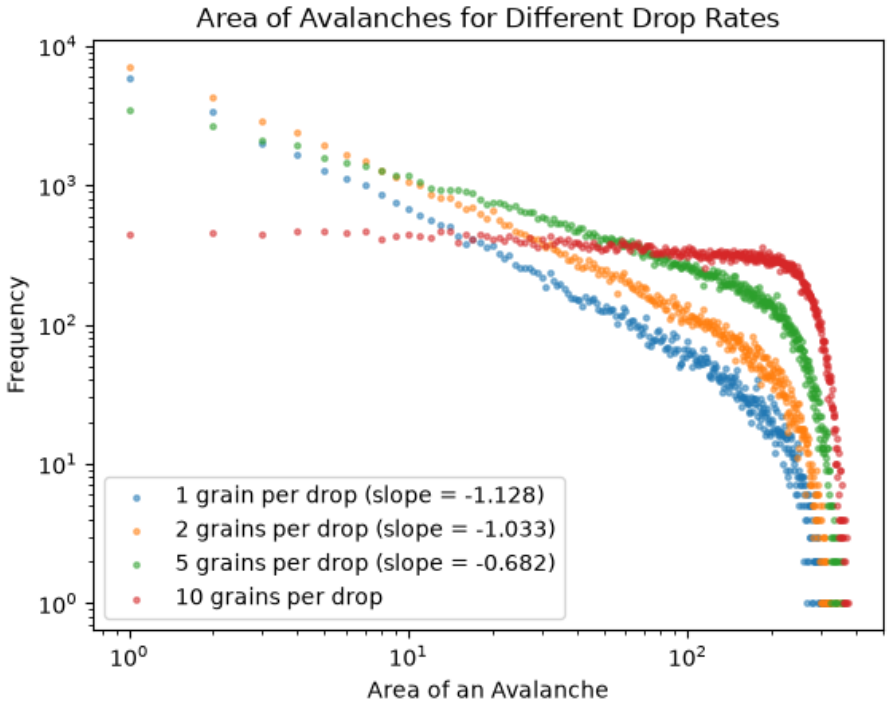


Figure 3.7: The area of avalanches for the extension model. The gradients of the graphs increase as the number of grains increases.

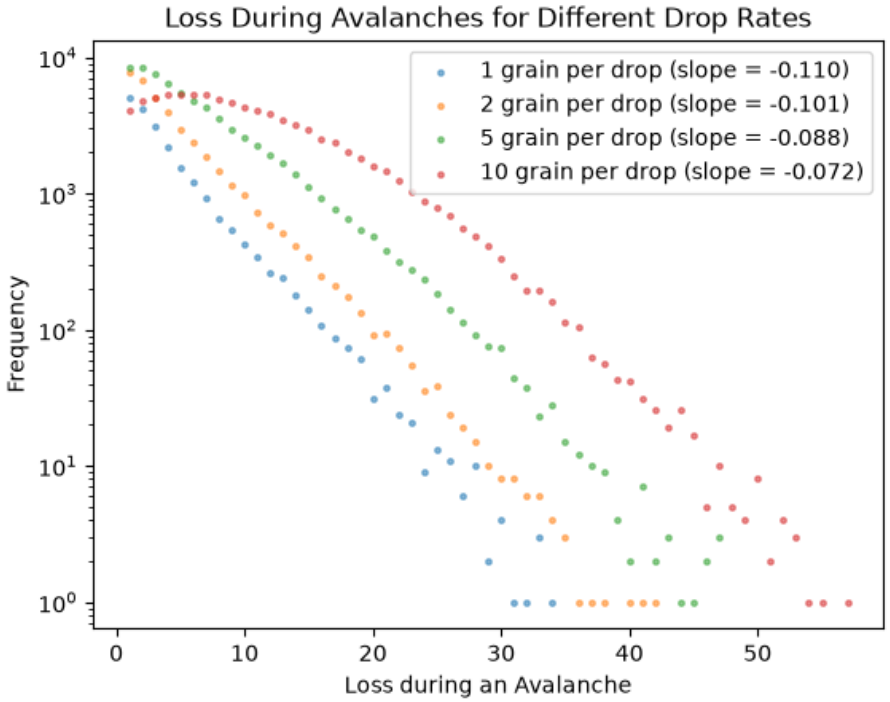


Figure 3.8: The loss for the extension model, with only frequency having a log axis. The graphs have linear segments, indicating there is an exponential relationship. For small numbers of loss, the graph is non-linear, especially so for the larger number of drops.

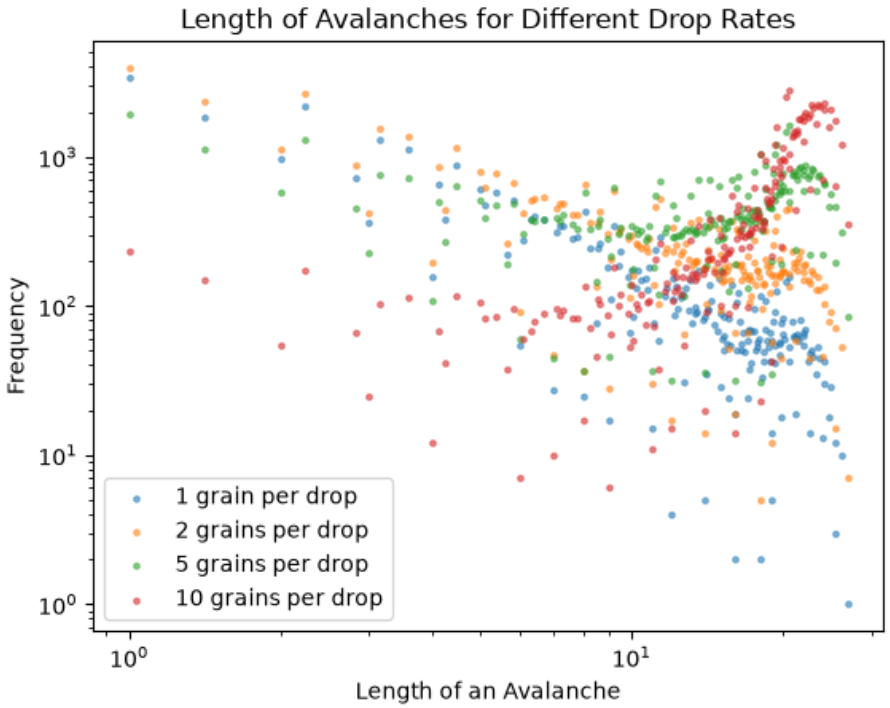


Figure 3.9: The length of avalanches for the extension model. The graphs are clearly non-linear when more grains are dropped.

3.3 Conclusion

To conclude, we see clear signs of SOC in the BTW model through the avalanche-topple and avalanche-area distributions. We also saw that the average height of the grid increased with grid size, approaching the known theoretical limit of 2.125. Further, all four avalanche statistics showed pairwise correlation. When multiple grains were added simultaneously, the topples and area distributions continued to exhibit power-law behaviour, although their fitted slows were varying. In contrast, the loss and length distributions varied as the number of grains added changed, highlighting how the way the system is driven can influence the form of these statistics. Overall, the results demonstrate how complex behaviour can arise from the simple BTW model.

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